

Evaluating Adaptive Differential Privacy for Federated IoHT Intrusion Detection with Heterogeneous Clients

Jess Alencaster

*Computer Science and
Software Engineering*

California Polytechnic State University
San Luis Obispo, CA, United States
jalencas@calpoly.edu

McCay Ruddick

*Computer Science and
Software Engineering*

California Polytechnic State University
San Luis Obispo, CA, United States
mruddick@calpoly.edu

Helen Huy

*Computer Science and
Software Engineering*

California Polytechnic State University
San Luis Obispo, CA, United States
hhuy@calpoly.edu

Abstract—Federated learning can reduce raw data sharing in Internet-of-Healthcare-Things (IoHT) intrusion detection, but client model updates may still leak training information. This work evaluates fixed and adaptive client-side differential privacy (DP) for federated IoHT intrusion detection using ECU-IoHT and WUSTL-EHMS 2020 with Dirichlet-based non-IID client splits. We compare no-DP, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping using IDS utility metrics and a white-box membership inference attack. Our main finding is that adaptive DP did not provide a clear advantage in this experimental setting. Strict no-identifier preprocessing reduced membership inference AUC close to chance, leaving little measurable leakage for DP to further reduce, while epsilon-scale noise large enough to provide stronger approximate privacy caused utility collapse. ECU-IoHT also remained near the utility ceiling, while WUSTL-EHMS exposed stronger noise sensitivity. Our results show that not every IoHT dataset or preprocessing approach is informative for demonstrating adaptive privacy gains. We find that adaptive mechanisms need both measurable leakage and utility tolerance for stronger perturbation to show benefit.

Index Terms—Federated learning, differential privacy, IoHT, intrusion detection, membership inference, non-IID data, privacy–utility tradeoff

I. INTRODUCTION

Internet-of-Healthcare-Things (IoHT) devices generate sensitive medical and network traffic data that can support machine learning-based intrusion detection. Centralizing this data raises privacy concerns, so federated learning (FL) is attractive because clients train locally and share model updates rather than raw records [1], [2]. However, model updates can still reveal information about local training data.

Differential privacy (DP) reduces update leakage by clipping updates and adding noise before aggregation. Many DP-FL methods use one fixed noise setting for all clients. IoHT clients may differ in data size, class balance, and traffic distribution. In healthcare settings, stronger privacy protection is especially important to protect sensitive patient data. However, intrusion detection also has to remain accurate enough to identify attacks. This creates a difficult privacy–utility tradeoff. Specifically, stronger DP noise can improve privacy protection,

but excessive noise may reduce detection performance, while weaker noise may preserve IDS utility but provide weaker formal privacy. Adaptive privacy mechanisms attempt to improve this tradeoff by adjusting noise or clipping across clients or rounds. Prior work has studied adaptive local DP in FL and IoT settings [6]–[8]. However, for heterogeneous IoHT intrusion-detection data, its practical benefit is still under-characterized.

In this work, we evaluate fixed and adaptive client-side privacy mechanisms on ECU-IoHT and WUSTL-EHMS 2020. We then simulate heterogeneous clients using Dirichlet-based partitioning, measure IDS utility using F1-score, and PR-AUC. Finally, we evaluate and report empirical privacy leakage using a white-box membership inference attack.

Our contributions are:

- We evaluate fixed and adaptive client-side privacy mechanisms under heterogeneous, non-IID federated IoHT clients.
- We compare ECU-IoHT and WUSTL-EHMS 2020 using both IDS utility metrics and white-box membership inference.
- We show that, under the tested attack and parameterization, the adaptive mechanisms did not reduce measured membership-inference leakage relative to fixed-noise perturbation, because strict preprocessing already leaves little measurable membership leakage, while noise large enough to target stronger nominal privacy scales substantially degrades utility, particularly on WUSTL-EHMS.

The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the methodology. Section IV describes the experiments and results. Section V concludes the paper.

II. RELATED WORK

FL has been widely used for distributed learning when raw data cannot be centralized [1]. In healthcare, FL is motivated by sensitive data and non-IID client populations [2]. Recent IoHT intrusion-detection work has also combined FL with

privacy mechanisms to protect healthcare network data during IDS training [10].

DP provides formal privacy protection by adding calibrated noise to a function output. The Gaussian mechanism gives approximate (ϵ, δ) -DP, with noise determined by sensitivity, ϵ , and δ [3], [4]. In deep learning, gradient clipping and Gaussian noise addition during training have been used to provide record-level DP guarantees with a formal composition accountant [15]. Local DP has been studied in federated IoT settings, where privacy budget and noise level directly affect utility [5]. Adaptive DP methods try to improve this balance by changing noise or clipping based on client or training behavior [6]–[8], including adaptively setting the clipping norm itself from the observed distribution of update norms [16]. Secure aggregation provides a complementary defense by preventing the server from observing individual client updates before aggregation [21].

Membership inference attacks evaluate whether a trained model reveals that a specific record was included in training [9]. This threat is important in federated learning. FL does reduce raw data sharing, but it does not eliminate leakage through model updates, gradients, or final model behavior. In IoHT intrusion detection, membership leakage is especially concerning because records may correspond to sensitive healthcare-device traffic that can involve patient data.

White-box membership inference provides a stronger evaluation setting than black-box attacks because the adversary can use internal model signals rather than only predicted labels or confidence scores [18]. These signals may include losses, gradients, logits, or update-derived features, all of which can expose differences between records seen during training and records held out from training. For this reason, white-box MIA is useful for evaluating whether privacy mechanisms reduce observable training-record leakage in the actual FL pipeline.

However, MIA results depend strongly on the dataset and preprocessing choices. Identifier-like features, duplicated patterns, or highly client-specific traffic can increase attack success because the model may learn shortcuts that separate training records from non-training records. Conversely, removing identifiers and other directly identifying fields can reduce attack signal before DP is even applied. This distinction is important because a low MIA AUC after preprocessing does not necessarily mean that DP caused the reduction. It instead may indicate that the preprocessing pipeline already removed much of the measurable membership signal.

Therefore, empirical MIA evaluation and formal DP analysis answer different questions. DP characterizes the theoretical protection provided by a randomized mechanism under assumptions about sensitivity, noise, composition, and privacy accounting. MIA measures whether a particular attacker can exploit the trained model or update signals under a specific dataset, preprocessing strategy, and training procedure. Evaluating both is necessary because a method may preserve utility and reduce empirical leakage while still providing weak formal privacy, or it may provide stronger formal privacy only at a utility cost that makes the IDS unusable.

This gap motivates our evaluation. Rather than assuming that adaptive DP will improve the privacy–utility tradeoff, we test whether adaptive noise scaling and adaptive clipping provide measurable benefit over fixed-noise DP under heterogeneous IoHT client splits. This allows us to separate three effects: first, whether the datasets contain measurable membership leakage; second, whether DP noise reduces that leakage; and third, whether adaptive mechanisms preserve IDS utility better than fixed privacy settings.

III. METHODOLOGY

A. System Model

We consider a federated IoHT intrusion-detection setting with K heterogeneous clients. Each client represents a local healthcare environment, such as a glucose-monitoring, blood-pressure monitoring, or patient-monitoring system, as indicated by Fig. 1.

Client k holds a private local dataset D_k , which may differ from other clients in size, class balance, and traffic distribution. Raw traffic remains local. In each communication round, clients train local IDS models, compute updates, apply the assigned privacy mechanism, and send only the resulting updates to the server. The server aggregates the updates using FedAvg and redistributes the updated global IDS model.

B. Threat Model

We consider the privacy threat that an adversary attempts to infer whether a record was used during local training. The adversary may observe model-related signals used by the white-box attack, such as model outputs, losses, gradients, or update-derived features, but does not directly access raw client records. This reflects a federated setting where raw IoHT traffic remains local, but shared model behavior may still reveal information about individual samples. The attacker’s goal is to distinguish member records from non-member records, and we measure success using membership inference ROC-AUC.

This manuscript does not formally define an adjacency relation or a protected unit (e.g., a single record, a single client, or one client’s participation across rounds) for the mechanisms in Section III-D. The membership-inference evaluation above tests record-level membership, while the clipping and noise mechanisms in Section III-D operate on an entire client update Δ_k^t . These two granularities are not necessarily aligned, and we do not claim a proven record-level or client-level (ϵ, δ) -DP guarantee; the approximate ϵ values reported in Section III-D and Section IV-G should be read as heuristic single-update noise-to-clipping scale comparisons rather than a certified end-to-end privacy guarantee.

C. Federated Learning Formulation

Let θ^t denote the global model parameters at round t . The server broadcasts θ^t , and each client trains locally to obtain θ_k^t . The client update is

$$\Delta_k^t = \theta_k^t - \theta^t.$$

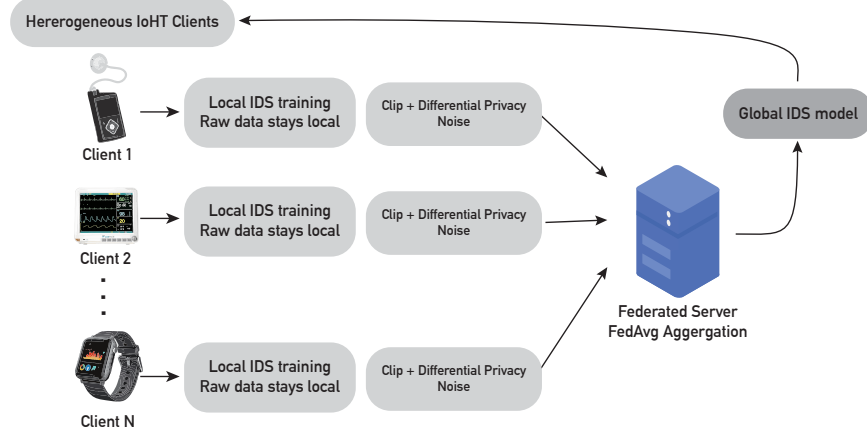


Fig. 1. System model for federated IoHT intrusion detection with heterogeneous clients. Raw data remain local to each IoHT environment, while privatized model updates are sent to the federated server for aggregation.

The server aggregates updates using FedAvg:

$$\theta^{t+1} = \theta^t + \sum_{k=1}^K w_k \tilde{\Delta}_k^t,$$

where w_k is the aggregation weight and $\tilde{\Delta}_k^t$ is the update received by the server. In the no-DP setting, $\tilde{\Delta}_k^t = \Delta_k^t$. In DP settings, the update is clipped and noised before upload.

D. Client-Side Privacy Mechanisms

For DP-based methods, each client update is clipped:

$$\bar{\Delta}_k^t = \Delta_k^t \cdot \min \left(1, \frac{C_t}{\|\Delta_k^t\|_2} \right),$$

where C_t is the clipping threshold. Gaussian noise is then added:

$$\tilde{\Delta}_k^t = \bar{\Delta}_k^t + \mathcal{N}(0, \sigma_{k,t}^2 C_t^2 I).$$

The Gaussian mechanism links noise, sensitivity, and privacy. For a function with ℓ_2 -sensitivity $\Delta_2(f)$, adding Gaussian noise with standard deviation

$$\sigma \geq \frac{c \Delta_2(f)}{\epsilon}, \quad c^2 > 2 \ln(1.25/\delta),$$

gives (ϵ, δ) -DP for $\epsilon \in (0, 1)$ [3], [4]. In our update mechanism, clipping sets the sensitivity scale through C_t , and the actual noise standard deviation is proportional to $\sigma_{k,t} C_t$. Thus, for fixed C_t and δ , increasing $\sigma_{k,t}$ decreases the approximate per-update ϵ , but also perturbs the IDS update more strongly. A full FL privacy budget would additionally depend on client sampling, number of rounds, composition, and the accountant used; here we report only approximate per-update privacy scales.

We note two further limitations of this privacy characterization. First, the bound above is classically stated for $\epsilon \in (0, 1)$ [3], [4]; the large approximate ϵ values reported for our utility-preserving noise settings (Section IV-G) fall well outside this range and should be read only as an informal noise-to-clipping scale comparison, not as a rigorous (ϵ, δ) -DP

certificate. An analytic Gaussian mechanism would be required to calibrate noise rigorously at these scales [17]. Second, the smoothed adaptive noise rule and the adaptive clipping rule described below select $\sigma_{k,t}$ or C_t using statistics of the private, unperturbed update Δ_k^t . We do not provide a privacy proof for this data-dependent parameter selection; standard Gaussian-mechanism accounting assumes the noise scale is fixed independently of the private input, and does not automatically extend to a mechanism whose variance or clipping threshold is itself a function of that input. Accordingly, we describe these two mechanisms as adaptive Gaussian perturbation and adaptive clipping heuristics rather than as mechanisms with a proven end-to-end DP guarantee.

We compare three DP variants. In fixed-noise DP, all clients use the same $\sigma_{k,t} = \sigma$. In smoothed adaptive noise scaling,

$$s_{k,t} = \lambda s_{k,t-1} + (1 - \lambda) \|\Delta_k^t\|_2,$$

$$\sigma_{k,t} = \text{clip} \left(\sigma_0 \left(1 + \beta \log \left(1 + \frac{s_{k,t}}{C_t} \right) \right), \sigma_{\min}, \sigma_{\max} \right).$$

In adaptive clipping, σ remains fixed while C_t changes during training; adapting the clipping norm to the observed distribution of update norms is a related idea explored in prior DP-FL work [16]. Table I summarizes these three mechanisms alongside the no-DP baseline.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Datasets and Preprocessing

We evaluate on ECU-IoHT and WUSTL-EHMS 2020. ECU-IoHT was introduced for analyzing cyberattacks in Internet of Health Things environments. ECU-IoHT is built from a controlled, simulated testbed. It contains eight to nine core features with over 111,207 total records. The simulated devices are connected through Wi-Fi, Bluetooth, and local subnets. Meanwhile, WUSTL-EHMS 2020 was created from real network traffic, including real physiological data captured from an Enhanced Healthcare Monitoring System testbed [11], [12]. It contains 44 features, including eight biometric and

TABLE I
SUMMARY OF EVALUATED FEDERATED PRIVACY MECHANISMS.

Method	Update mechanism	Adaptive behavior	Role in evaluation
No-DP FL	Clients send the unmodified local update Δ_k^t .	None.	Utility and leakage baseline.
Fixed-noise DP	Clients send clipped updates with Gaussian noise using fixed σ and C .	None; all clients use the same privacy setting.	Uniform DP baseline.
Smoothed adaptive noise	Clients send clipped updates with client- and round-specific noise $\sigma_{k,t}$.	Noise changes using a smoothed update-norm statistic.	Tests adaptive noise scaling under heterogeneous clients.
Adaptive clipping	Clients send noised updates after applying a changing clipping threshold C_t .	Clipping changes during training while noise remains fixed.	Tests adaptive sensitivity control.

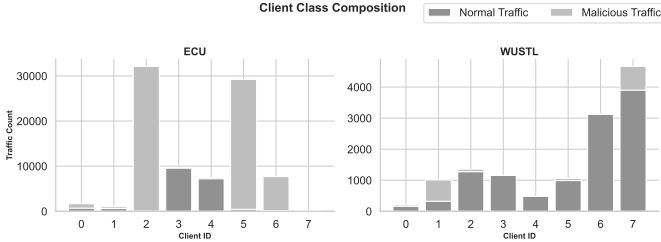


Fig. 2. Dirichlet-based federated client split for ECU-IoHT and WUSTL-EHMS 2020 using $K = 8$ clients and $\alpha = 0.3$. The split creates heterogeneous client sizes and class compositions.

physiological features such as pulse rate, blood pressure, and oxygen saturation. Both have been used in privacy-preserving FL-based IoHT IDS work [10].

For the main evaluation, we remove identifier-like fields including IDs, MAC-like features, and direct identifiers. We also use an identifier-inclusive WUSTL condition as a positive-control leakage condition to confirm that the membership inference attack detects leakage when leaky features are retained.

B. Federated Client Partitioning

The datasets are centralized, so we simulate federated clients using label-based Dirichlet partitioning:

$$p \sim \text{Dirichlet}(\alpha).$$

Smaller α values create more uneven client label distributions. This matches common non-IID FL evaluation practice; the FedDC supplemental uses $\alpha = 0.3$ and $\alpha = 0.6$, with 0.3 representing the stronger non-IID setting [13], [14]. We use $K = 8$ clients and $\alpha = 0.3$ to create heterogeneous client sizes and class balances. Fig. 2 shows the resulting split. We note that this split creates label-distribution heterogeneity across clients; it does not by itself model device-type or feature-distribution heterogeneity (e.g., differences between glucose-monitoring, blood-pressure, and other IoHT device types shown in Fig. 1), which we leave to future work.

C. Experimental Parameters

Table II lists the main privacy-related values needed to interpret the privacy-utility experiments, including the noise multiplier σ , clipping norm C , sensitivity scale, and target ϵ , δ . We report ϵ only as an approximate per-update scale

TABLE II
MAIN EXPERIMENTAL PARAMETERS.

Parameter	Value
Clients	$K = 8$
Client split	Dirichlet label split, $\alpha = 0.3$
Fixed DP noise	$\sigma = 0.010$
Adaptive sigma base	$\sigma_0 = 0.010$
Adaptive sigma rule	$\lambda = 0.8, \beta = 0.15$
Sigma bounds	$[0.001, 0.10]$
Fixed clip norm	$C = 1.0$
Noise sweep	0.005–1.0
Target epsilon sensitivity	$\epsilon = 2\text{--}8, \delta = 10^{-5}$
Privacy accounting	Approx. per-update only

(Section III-D); a composed training-level budget would additionally require the number of FL rounds, the client sampling rate, and a formal accountant, which are not computed in this version of the manuscript.

D. Training Conditions and Metrics

We compare no-DP training, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping. All conditions use the same client splits and FL pipeline so that differences are attributable to the privacy mechanism.

We evaluate IDS utility using accuracy, precision, recall, F1-score, and PR-AUC. Because the datasets are imbalanced, we emphasize F1-score and PR-AUC as our utility metrics. We evaluate privacy leakage separately using white-box membership inference ROC-AUC, where 0.5 indicates chance-level attack performance.

E. Privacy Evaluation and Budget Sensitivity

Membership inference is used as an empirical leakage test. Member samples come from records used to train the target model, and non-member samples come from held-out records. We report attack ROC-AUC, where 0.5 indicates chance-level performance. We evaluate the attack mainly after identifier removal so that the measured leakage reflects model behavior rather than direct identifier leakage; the identifier-inclusive WUSTL condition is used only as a positive control to confirm that the attack detects leakage when leaky features are present.

The privacy scale also depends on more than the noise multiplier alone. Under the Gaussian-mechanism estimate, clipping controls the sensitivity scale through C , while the added noise is proportional to σC . For fixed δ , larger σ lowers

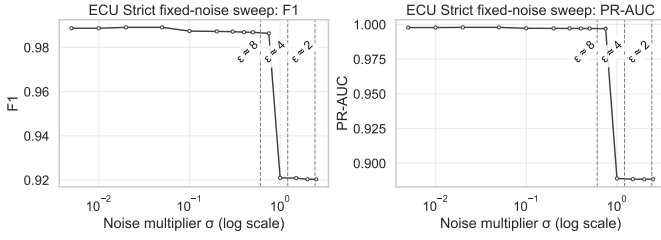


Fig. 3. Fixed-noise utility sweep for ECU-IoHT under preprocessing with identifiers removed. F1-score and PR-AUC remain high across the tested noise multipliers, showing limited room for adaptive utility gains.

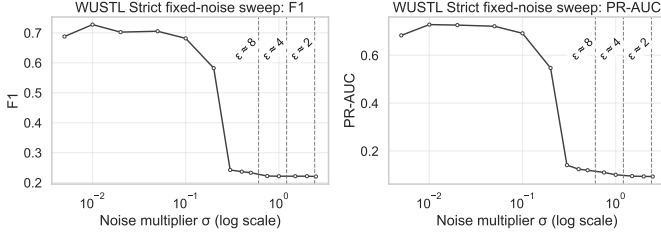


Fig. 4. Fixed-noise utility sweep for WUSTL-EHMS 2020 under preprocessing with identifiers removed. F1-score and PR-AUC are lower and more sensitive to the noise multiplier.

approximate per-update ϵ , but also perturbs the client update more strongly. A full training-level privacy budget would also require composition across FL rounds and client sampling, which we do not compute here. Therefore, we interpret ϵ only as an approximate per-update scale used to compare noise settings.

F. Fixed-Noise Utility Sweep

We first vary the fixed Gaussian noise multiplier under strict no-identifier preprocessing. This tests whether each dataset can tolerate the noise needed for DP.

Fig. 3 shows that ECU-IoHT remains near the utility ceiling across the tested noise range. Since F1-score and PR-AUC are already high, ECU-IoHT provides little room for adaptive mechanisms to show utility improvement.

Fig. 4 shows that WUSTL-EHMS 2020 is more sensitive: F1-score and PR-AUC are lower and degrade more as σ increases. This makes WUSTL more informative for observing the utility cost of privacy noise.

G. Membership Inference Results

Fig. 5 shows the white-box membership inference ROC-AUC under strict preprocessing with identifiers removed. For both datasets, attack performance stays near the chance line across no-DP, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping.

This is the key privacy result: after identifier removal, no-DP training already has little measurable membership leakage under our attack. Therefore, DP and adaptive DP have little empirical MIA signal left to reduce.

We report attack ROC-AUC only. This aggregate metric can understate leakage that is concentrated in a small subset of

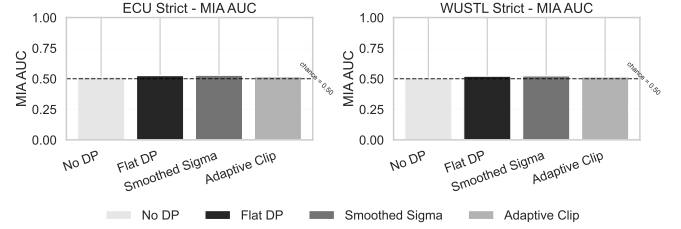


Fig. 5. White-box membership inference ROC-AUC under preprocessing with identifiers removed. Attack performance remains close to chance for both ECU-IoHT and WUSTL-EHMS 2020 across all evaluated privacy mechanisms.

records; true-positive rate at low false-positive rate [19] or a broader empirical evaluation protocol [20] would provide a stricter, complementary audit of the near-chance result reported here.

H. Heuristic Privacy Scale Comparison

Table III summarizes the approximate relationship between the tested noise multipliers and per-update Gaussian-mechanism privacy scale. As discussed in Section III-D, this scale is a heuristic comparison rather than a certified (ϵ, δ) -DP guarantee: the underlying bound is classically stated for $\epsilon \in (0, 1)$, does not account for composition across FL rounds or client sampling, and does not account for the data-dependent parameter selection used by the adaptive mechanisms. With $C' = 1.0$ and $\delta = 10^{-5}$, the utility-preserving settings use very small noise multipliers around $\sigma \approx 0.010$. These values preserve IDS performance but correspond to very large approximate ϵ on this scale, meaning weak formal privacy.

Targeting more conventional $\epsilon = 2\text{--}8$ on this same scale required much larger noise multipliers, approximately 0.43–1.65. These values are in the high-noise range where WUSTL utility degrades in Fig. 4. Thus, the experiment reveals the tradeoff that the noise needed for a stronger nominal privacy scale is not utility-tolerable in this setting, while lower noise preserves utility but gives a weak privacy scale.

We also note that the realized adaptive noise multiplier changed only marginally relative to the fixed baseline: Table III reports a mean $\sigma \approx 0.011$ for smoothed adaptive noise versus $\sigma = 0.010$ for fixed noise, and Table II caps the adaptive multiplier at $\sigma_{\max} = 0.10$, which remains below the noise multipliers associated with the target $\epsilon = 2\text{--}8$ scale above. The tested adaptive rule therefore explored a narrow, low-noise operating region rather than the full range spanned by the fixed-noise sweep; this experiment should be read as evaluating adaptation near a low-noise baseline rather than adaptation across the complete privacy–utility range.

I. Interpretation

The tested adaptive mechanisms did not show a clear advantage over fixed-noise perturbation under our attack and parameterization, because the evaluated datasets and the realized adaptive noise range (Section IV-G) leave little room for

TABLE III
NOISE MULTIPLIER AND HEURISTIC SINGLE-UPDATE NOISE-TO-CLIPPING
SCALE COMPARISON, COMPUTED FROM THE CLASSICAL
GAUSSIAN-MECHANISM BOUND IN SECTION III-D AT $\delta = 10^{-5}$. VALUES
ARE NOT COMPOSED ACROSS FL ROUNDS AND ARE NOT A CERTIFIED
(ϵ, δ)-DP GUARANTEE (SECTION III-D).

Method	Mean σ	Approx. ϵ
Fixed noise	0.010	484.5
Smoothed sigma	0.011	441–458
Adaptive clip	0.010	484.5
Target $\epsilon = 2-8$	$\approx 0.43-1.65$	2–8

adaptation to help. First, strict preprocessing removed most measurable membership leakage, so MIA AUC stayed near chance even without DP. Second, the noise multipliers that preserved utility produced weak approximate privacy scales. Third, increasing noise toward the $\epsilon = 2-8$ scale moved into the range where WUSTL utility degraded substantially, while ECU-IoHT remained near its utility ceiling across the same range (Fig. 3).

This means the result is not simply that adaptive DP failed or that the datasets were “too easy.” Rather, these datasets and preprocessing choices created a regime where privacy leakage was already empirically low, but stronger formal-noise settings were too damaging for WUSTL-EHMS utility. Under these conditions, the tested adaptive mechanisms can match fixed-noise perturbation but have limited opportunity to improve the privacy–utility tradeoff.

These results do not establish that adaptive privacy mechanisms are generally ineffective for federated IoHT intrusion detection. They instead identify a specific combination of preprocessing, attack, and adaptation range under which an evaluation of adaptive DP is uninformative: little leakage remained for adaptive noise to reduce, and the tested adaptation rule (Section IV-G) did not explore a privacy-relevant noise range.

V. CONCLUSION

This work evaluated fixed and adaptive client-side DP for federated IoHT intrusion detection under heterogeneous non-IID clients. Under the tested attack, preprocessing, and parameterization, the adaptive mechanisms did not reduce measured membership-inference leakage relative to fixed-noise perturbation, and did not produce a clear privacy or utility advantage in the evaluated regime.

The main limitation of this study is that strict no-identifier preprocessing reduced MIA AUC close to chance, leaving little measurable membership leakage for DP to reduce. In addition, the noise levels that preserved utility were too small to provide strong approximate privacy, while targeting $\epsilon = 2-8$ required much larger noise and caused substantial utility degradation on WUSTL-EHMS.

This study has additional limitations. We do not provide a composed, training-level (ϵ, δ)-DP guarantee across FL rounds, and the smoothed adaptive noise and adaptive clipping rules select their parameters from private, unperturbed update

statistics without a corresponding privacy proof for that selection step; the reported approximate ϵ values should therefore be read as heuristic scale comparisons rather than certified guarantees (Section III-D). We additionally do not report variability across multiple training seeds or independently sampled Dirichlet partitions, so small differences between mechanisms should not be interpreted as statistically significant.

Future work should focus on drift-based adaptive clipping, client-specific update diagnostics, and settings where privacy improvement can be measured directly. In particular, future experiments should use datasets or client splits with observable post-cleaning membership leakage and enough utility tolerance for meaningful DP noise, allowing adaptive privacy to be tested against fixed-noise DP without immediate utility collapse.

VI. ETHICAL CONSIDERATIONS

This work aims to improve privacy-preserving IoHT intrusion detection. We keep raw records local, remove identifier-like fields, and report aggregate results. The membership inference attack is used only to evaluate privacy leakage.

ACKNOWLEDGMENT

The authors thank the Computer Science and Software Engineering Department at California Polytechnic State University for supporting this work.

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. AISTATS*, 2017.
- [2] Prayitno, C.-R. Shyu, K. T. Putra, H.-C. Chen, Y.-Y. Tsai, K. S. M. T. Hossain, W. Jiang, and Z.-Y. Shae, “A systematic review of federated learning in the healthcare area: From the perspective of data properties and applications,” *Applied Sciences*, vol. 11, no. 23, Art. no. 11191, 2021.
- [3] C. Dwork, K. Kenthapadi, F. McSherry, I. Mironov, and M. Naor, “Our data, ourselves: Privacy via distributed noise generation,” in *Advances in Cryptology – EUROCRYPT 2006*, ser. Lecture Notes in Computer Science, vol. 4004. Springer, 2006, pp. 486–503.
- [4] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.
- [5] Y. Zhao, J. Zhao, M. Yang, T. Wang, N. Wang, L. Lyu, D. Niyato, and K.-Y. Lam, “Local differential privacy-based federated learning for Internet of Things,” *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 8836–8853, 2021.
- [6] G. Yue, L. Yan, L. Kang, and C. Shen, “AdapLDP-FL: An adaptive local differential privacy for federated learning,” *IEEE Transactions on Mobile Computing*, 2025, doi: 10.1109/TMC.2025.3533090.
- [7] Z. He, L. Wang, and Z. Cai, “Clustered federated learning with adaptive local differential privacy on heterogeneous IoT data,” *IEEE Internet of Things Journal*, vol. 11, no. 1, pp. 137–146, 2024.
- [8] M. Talaei and I. Izadi, “Adaptive differential privacy in federated learning: A priority-based approach,” arXiv:2401.02453, 2024.
- [9] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in *Proc. IEEE Symposium on Security and Privacy*, 2017.
- [10] F. Mosaiyebzadeh, S. Pouriyeh, M. Han, L. Liu, Y. Xie, L. Zhao, and D. M. Batista, “Privacy-preserving federated learning-based intrusion detection system for IoHT devices,” *Electronics*, vol. 14, no. 1, Art. no. 67, 2025.
- [11] M. Ahmed, S. Byreddy, A. Nutakki, L. F. Sikos, and P. Haskell-Dowland, “ECU-IoHT: A dataset for analyzing cyberattacks in Internet of Health Things,” *Ad Hoc Networks*, vol. 122, Art. no. 102621, 2021.

- [12] Washington University in St. Louis, “WUSTL-EHMS-2020 dataset for Internet of Medical Things cybersecurity research,” 2020. [Online]. Available: <https://www.cse.wustl.edu/~jain/ehms/index.html>
- [13] L. Gao, H. Fu, L. Li, Y. Chen, M. Xu, and C.-Z. Xu, “FedDC: Federated learning with non-IID data via local drift decoupling and correction,” in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10112–10121.
- [14] L. Gao, H. Fu, L. Li, Y. Chen, M. Xu, and C.-Z. Xu, “Supplementary for ‘FedDC: Federated Learning with Non-IID Data via Local Drift Decoupling and Correction’,” Supplementary material for CVPR 2022, 2022.
- [15] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, “Deep learning with differential privacy,” in *Proc. ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2016, pp. 308–318.
- [16] G. Andrew, O. Thakkar, H. B. McMahan, and S. Ramaswamy, “Differentially private learning with adaptive clipping,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 17455–17466.
- [17] B. Balle and Y.-X. Wang, “Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising,” in *Proc. International Conference on Machine Learning (ICML)*, 2018, pp. 394–403.
- [18] M. Nasr, R. Shokri, and A. Houmansadr, “Comprehensive privacy analysis of deep learning: Passive and active white-box inference attacks against centralized and federated learning,” in *Proc. IEEE Symposium on Security and Privacy*, 2019, pp. 739–753.
- [19] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in *Proc. IEEE Symposium on Security and Privacy*, 2022, pp. 1897–1914.
- [20] B. Jayaraman and D. Evans, “Evaluating differentially private machine learning in practice,” in *Proc. USENIX Security Symposium*, 2019, pp. 1895–1912.
- [21] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth, “Practical secure aggregation for privacy-preserving machine learning,” in *Proc. ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2017, pp. 1175–1191.